

Quantum Mechanics

Lecture 9: Free Particles, Fourier Evolution, and the Classical Limit

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Chapter 1

Free Particles, Fourier Evolution, and the Classical Limit

1.1 When Classical Repair Was No Longer Enough

There are two ways to respond when an old theory fails. One can try to repair the mechanism, adding enough machinery to preserve the familiar picture, or one can search for a principle that changes the picture itself. The Michelson–Morley experiment encouraged both reactions. Elaborate mechanical models of the ether could be made to imitate the observed result, but Einstein’s relativity began from a sharper principle and made the old mechanism unnecessary. Quantum theory followed a similarly reluctant path from repair to reconstruction.

For the present discussion, the decisive classical failure is blackbody radiation. Electromagnetic radiation in a cavity can be decomposed into normal modes. Each mode behaves mathematically like a harmonic oscillator, and classical equipartition assigns to it an average thermal energy of order

$$E_{\text{mode}} = kT. \quad (1.1)$$

The cavity contains infinitely many modes, extending to arbitrarily high frequency. If every one carries the nonzero energy in Equation (1.1), their sum diverges. The prediction is not merely somewhat inaccurate in the ultraviolet; it is infinite. This is the ultraviolet catastrophe.

Planck found the rule that cuts off the divergence:

$$E_n = nh\nu, \quad n = 0, 1, 2, \dots \quad (1.2)$$

At first, however, the rule was attached to oscillators in the cavity walls. The radiation itself was still imagined as an ordinary classical wave. Einstein took the more radical step and attributed the quanta to the radiation. If $h\nu \gg kT$, thermal agitation ordinarily cannot supply even one quantum at that frequency. A classical fraction of a quantum is not available; the most probable occupation is zero. High-frequency modes are therefore suppressed.

This did not make photons immediately welcome. A wave theory had explained interference and diffraction too well to be discarded casually. The right conclusion was not that waves were simply false, but that the classical wave description was incomplete. Compton scattering eventually made the particle-like transfer of energy and momentum especially difficult to evade.

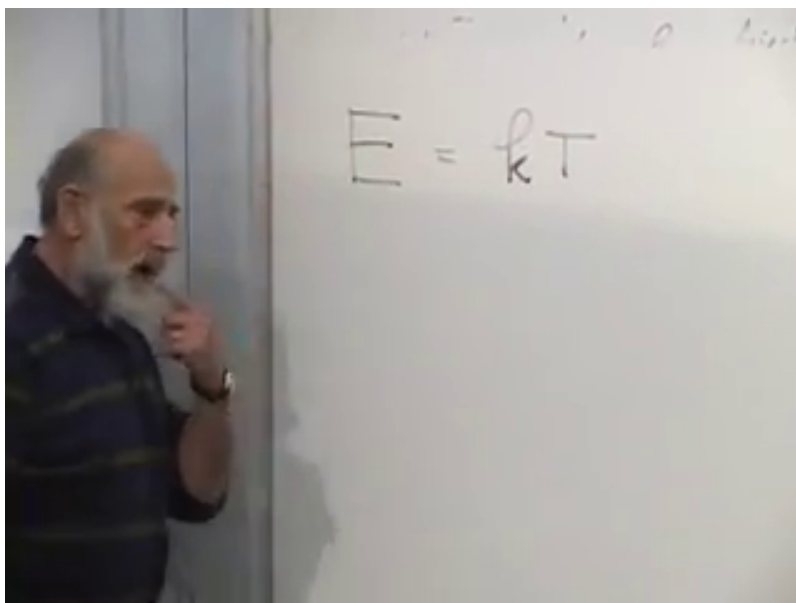


Figure 1.1: The classical equipartition assignment for each electromagnetic normal mode of a thermal cavity.

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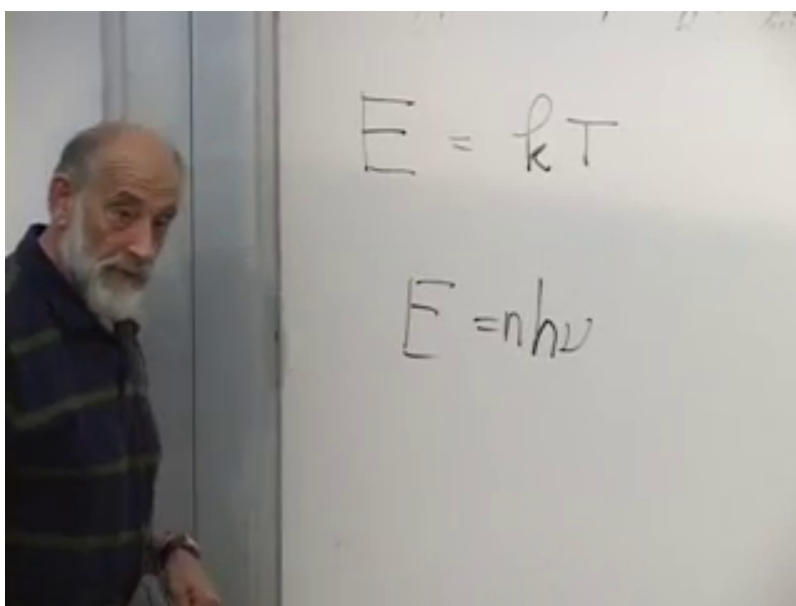


Figure 1.2: Planck's discrete energy rule, whose interpretation Einstein extended from material emitters to radiation itself.

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Question from the audience. Why was the photoelectric effect not already decisive evidence for photons? ^a

Response. It was important evidence, and Einstein's explanation was one of the earliest direct uses of light quanta. Historically, however, it did not force everyone to abandon a classical field interacting with quantized matter. Compton scattering displayed photon-like energy and momentum conservation so transparently that the remaining classical interpretations became much harder to sustain.

^aLecture timestamp: 00:10:09–00:10:19.

Question from the audience. Where does Bohr's 1913 atomic model belong in this transition? ^a

Response. It is one of several sharp breaks rather than the final theory. Bohr imposed discrete stationary orbits and quantum jumps on a largely classical orbital picture. The complete conceptual transition came later, especially through Heisenberg's matrix mechanics, Schrödinger's wave mechanics, and Dirac's unifying operator language.

^aLecture timestamp: 00:10:20–00:11:13.

Einstein had also recognized very early that atomic and radioactive events carry an irreducibly statistical appearance. An excited atom can remain undecayed for an unpredictable interval and then emit spontaneously. A radioactive nucleus behaves similarly. One can predict distributions and lifetimes with great precision without predicting the moment of an individual event.

The old orbital models tried to preserve familiar mechanics by adding special rules: only certain orbits were allowed, and radiation somehow occurred only in jumps between them. Such models could organize data, but they did not provide a consistent mechanics. Heisenberg's decisive move was to stop describing unobservable electron orbits and build the theory from observable transition quantities.

Question from the audience. Why did the old orbit models persist if they were already in tension with classical mechanics? ^a

Response. They retained a familiar picture and reproduced important spectral regularities. That made them useful transitional models. Their special rules were not a coherent dynamics, however, and increasingly complicated atoms exposed the weakness. Heisenberg broke with the orbit picture rather than adding another repair.

^aLecture timestamp: 00:12:50–00:13:22.

Question from the audience. If every observation of a photon uses a quantized charged particle, could quantizing only the matter be enough? ^a

Response. Not in a consistent interaction theory. A quantum system entangles with whatever it exchanges energy and momentum with. Leaving the electromagnetic field classical while matter is quantum can imitate limited experiments, but it cannot consistently describe emission, absorption, fluctuations, and photon statistics together. Both sides of the interaction must ultimately be treated quantum mechanically.

^aLecture timestamp: 00:14:06–00:14:55.

Question from the audience. Why did Einstein later resist quantum mechanics after recognizing spontaneous probability so early? ^a

Response. He recognized the empirical need for probabilistic transition laws, but he remained dissatisfied with probability as the final description of an individual physical system. The issue was not whether the statistical predictions worked; it was what they meant and whether a deeper state of affairs had been omitted. His later objections were attempts to show that the formalism, though successful, might not be complete.

^aLecture timestamp: 00:14:59–00:17:08.

Question from the audience. Could probability merely reflect ignorance, as it does when a classical coin is tossed? ^a

Response. That possibility was precisely the source of the dispute. Quantum mechanics assigns probabilities even when the state has been specified as completely as the theory allows. The formalism predicts not only uncertainty but interference between alternatives. Treating those probabilities as ordinary ignorance does not reproduce the full structure without adding a much more subtle underlying theory.

^aLecture timestamp: 00:17:09–00:18:19.

The historical prelude fixes the direction of the argument that follows. Quantum mechanics is the primary framework. Classical mechanics is not used to derive it; at most, classical expressions suggest a Hamiltonian whose quantum consequences we then work out. Finally we must recover classical motion as an approximation to the quantum theory.

1.2 The Free Hamiltonian as a Quantum Guess

The abstract Schrödinger equation is

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle. \quad (1.3)$$

In the position representation, momentum is

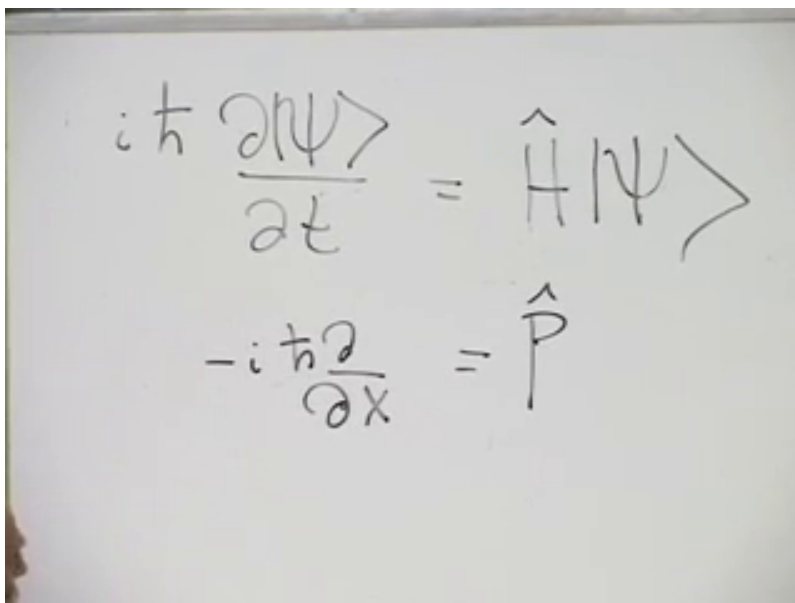
$$\hat{p} = -i\hbar \frac{\partial}{\partial x}. \quad (1.4)$$

The two equations have a revealing parallel. A derivative with respect to time is generated by the Hamiltonian; a derivative with respect to position is generated by momentum. The sign in Equation (1.4) is conventional, but once chosen it must be used consistently.

For a nonrelativistic free particle on a line, we guess

$$\boxed{\hat{H} = \frac{\hat{p}^2}{2m}}. \quad (1.5)$$

This is a correspondence-guided guess, not a derivation of quantum mechanics from classical mechanics. Free space has no preferred direction, so the energy should be unchanged under $p \mapsto -p$. The simplest even function with the familiar low-speed behavior is $p^2/(2m)$. The constant m earns its interpretation as inertial mass through the resulting relation between momentum and velocity.

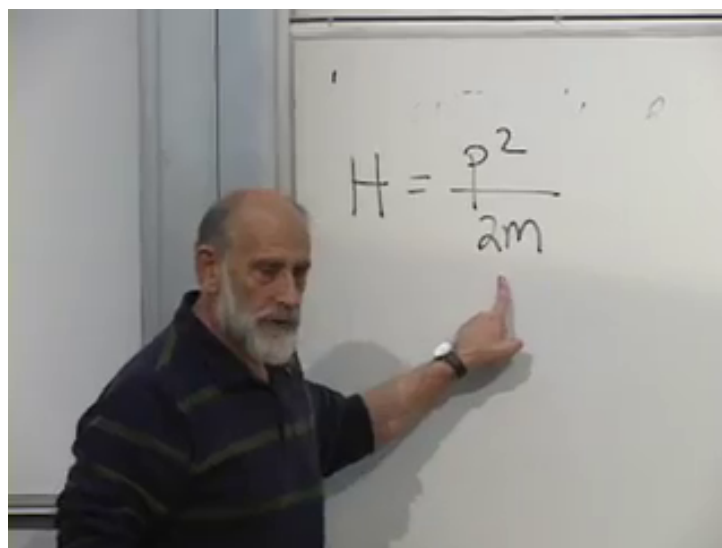


$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}|\psi\rangle$$

$$-i\hbar \frac{\partial}{\partial x} = \hat{P}$$

Figure 1.3: Time and space derivatives paired with the Hamiltonian and momentum operators.

Lecture frame: 00:21:40.



$$H = \frac{p^2}{2m}$$

Figure 1.4: The nonrelativistic free-particle Hamiltonian chosen as an even function of momentum.

Lecture frame: 00:24:00.

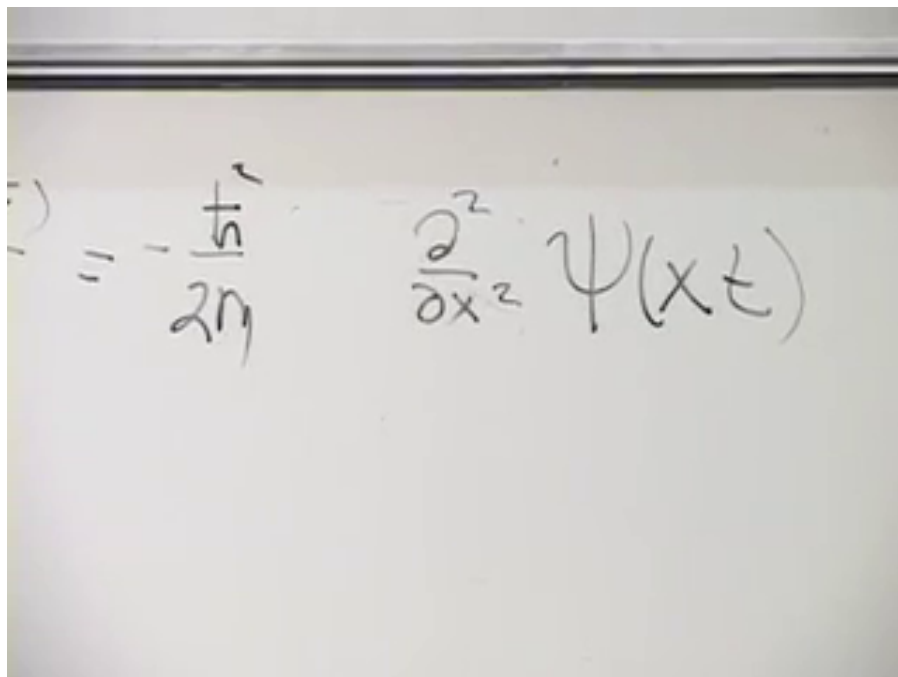


Figure 1.5: The free Schrödinger equation obtained by applying the momentum operator twice.

Lecture frame: 00:28:02.

The square of an operator means composition: $\hat{p}^2\psi$ is $\hat{p}(\hat{p}\psi)$. Writing $\psi(x, t) = \langle x | \psi(t) \rangle$, the probability of finding the particle in an interval dx at a specified time is

$$\Pr(x \in [x, x + dx]; t) = |\psi(x, t)|^2 dx. \quad (1.6)$$

Position labels the possible outcomes of an observable. Time labels the evolution of the state; in this elementary formulation it is not another coordinate whose value is measured with probability $|\psi|^2$.

Substituting Equations (1.4) and (1.5) into Equation (1.3) gives

$$i\hbar \frac{\partial \psi}{\partial t} = \frac{1}{2m} \left(-i\hbar \frac{\partial}{\partial x} \right)^2 \psi \quad (1.7)$$

$$= -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2}. \quad (1.8)$$

Thus the free-particle equation is

$$\boxed{i\hbar \frac{\partial \psi(x, t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x, t)}{\partial x^2}}. \quad (1.9)$$

1.3 Plane Waves and the Two Directions of Momentum

We first seek a solution with definite momentum. Following the sign convention used at the board, take

$$\psi(x, t) = e^{ipx/\hbar} e^{i\omega t}. \quad (1.10)$$

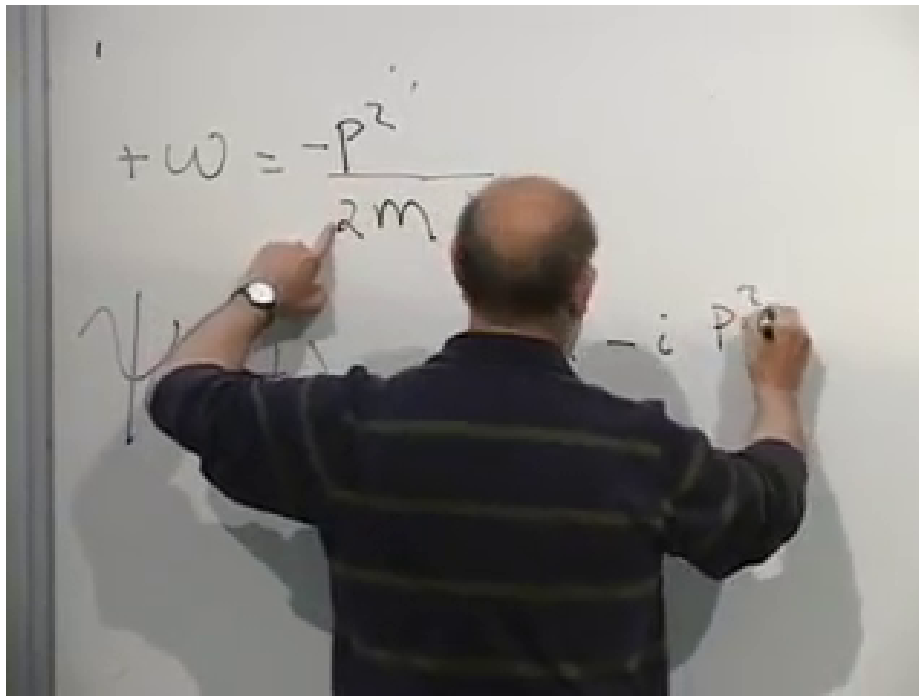


Figure 1.6: The plane-wave substitution fixes the time frequency through $E = p^2/(2m)$.

Lecture frame: 00:34:45.

Question from the audience. Should the frequency ω also appear in the spatial exponent? ^a

Response. No. The spatial exponent is fixed by the momentum eigenvalue: $-i\hbar\partial_x e^{ipx/\hbar} = p e^{ipx/\hbar}$. The independent time factor is then determined by the Schrödinger equation. If one writes the wave as $e^{i(kx-\omega t)}$, the relation is $p = \hbar k$ and $E = \hbar\omega$.

^aLecture timestamp: 00:29:05–00:29:20.

The time derivative of Equation (1.10) is $\partial_t \psi = i\omega \psi$, while

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{p^2}{\hbar^2} \psi. \quad (1.11)$$

The free Schrödinger equation therefore requires

$$-\hbar\omega = \frac{p^2}{2m}, \quad \omega = -\frac{p^2}{2m\hbar}. \quad (1.12)$$

The negative value is a consequence of having written the time factor as $e^{+i\omega t}$. In the conventional positive-frequency notation,

$$\psi_p(x, t) = \exp \left[\frac{i}{\hbar} \left(px - \frac{p^2}{2m} t \right) \right], \quad (1.13)$$

so $E = p^2/(2m)$ and the time factor is $e^{-iEt/\hbar}$.

The same energy belongs to two momentum eigenstates:

$$p_E = \pm \sqrt{2mE}. \quad (1.14)$$

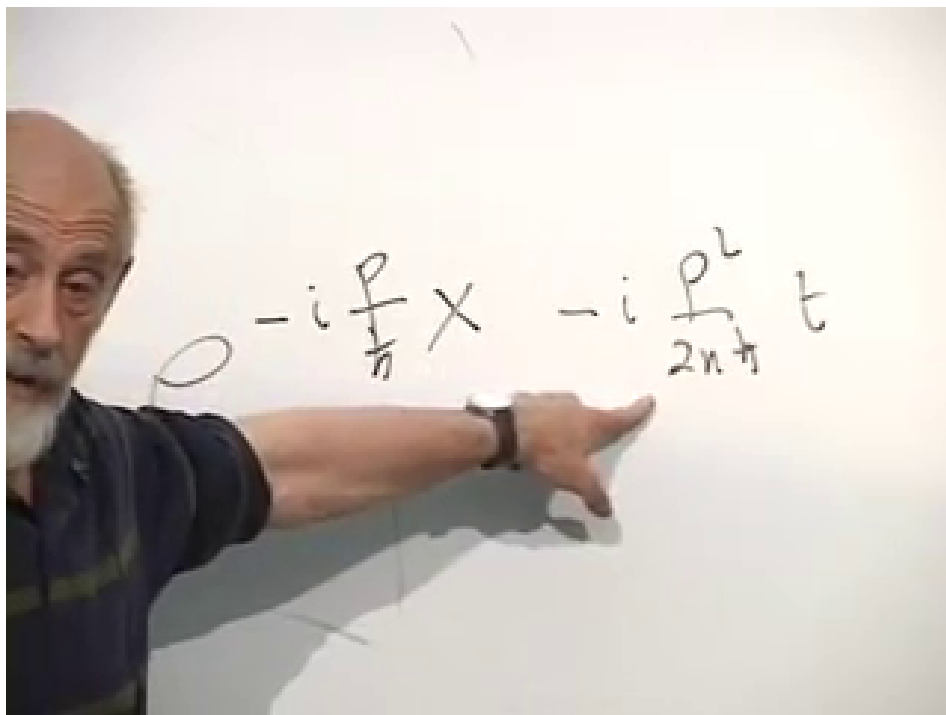


Figure 1.7: The $+p$ and $-p$ plane waves have a common energy because the Hamiltonian depends on p^2 .

Lecture frame: 00:37:40.

The right-moving and left-moving waves differ in their spatial phases but share the same time phase. Any superposition

$$\psi(x, t) = \alpha\psi_p(x, t) + \beta\psi_{-p}(x, t) \quad (1.15)$$

therefore remains in the same energy eigenspace. In a two-state, box-normalized version of the problem, the probabilities for momenta $\pm p$ are $|\alpha|^2$ and $|\beta|^2$, while the probability of the common energy is their sum.

Ideal plane waves are not square-integrable on the whole line. They are generalized, delta-normalized eigenstates. Normalizable particle states are wave packets assembled from them.

1.4 The General Free State in Momentum Space

Linearity now gives the general free evolution as a Fourier superposition:

$$\boxed{\psi(x, t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp e^{ipx/\hbar} e^{-ip^2 t/(2m\hbar)} \tilde{\psi}(p).} \quad (1.16)$$

The coefficient $\tilde{\psi}(p)$ is the momentum-space amplitude at $t = 0$. Every value of momentum evolves independently by the phase dictated by its energy.

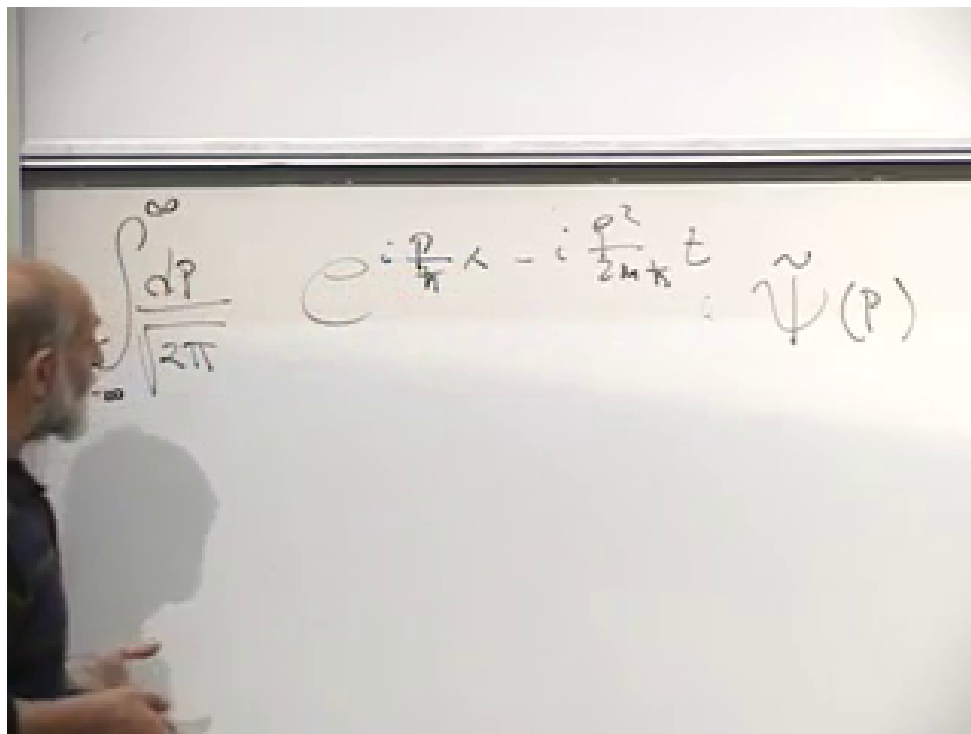


Figure 1.8: The general free state as a continuous superposition of momentum eigenwaves.

Lecture frame: 00:41:10.

Question from the audience. In the integral, is p a particular measured momentum or only a dummy variable? ^a

Response. Inside the integral it is a dummy variable ranging over all real momenta. A particular outcome may instead be denoted q . Its probability density is $|\tilde{\psi}(q)|^2$, independent of what symbol was used for the integration variable.

^aLecture timestamp: 00:41:55–00:42:13.

Question from the audience. What is held fixed while the momentum integral is performed? ^a

Response. The spacetime point x, t is fixed. We sum the complex contribution made at that point by every momentum component. Constants such as m and $\hbar$ are fixed as well; only the integration variable runs.

^aLecture timestamp: 00:42:14–00:42:57.

With the normalization of Equation (1.16),

$$\Pr(p \in [q, q + dq]) = |\tilde{\psi}(q)|^2 dq. \quad (1.17)$$

Question from the audience. Does the symbol q in $P(q)$ denote a position? ^a

Response. No. Here q is merely a fresh name for a possible momentum outcome. Position is still denoted x . The change of letter prevents confusion with the dummy momentum p already used inside the integral.

^aLecture timestamp: 00:44:26–00:45:05.

Figure 1.9: The momentum-space Born rule: $P(q) = \tilde{\psi}^*(q)\tilde{\psi}(q)$.

Lecture frame: 00:44:24.

Energy is a coarser label than momentum because the map

$$E = \frac{p^2}{2m} \quad (1.18)$$

identifies p and $-p$.

For discrete or box-normalized momentum states, the weight assigned to energy E is the sum of the weights at its two roots. For a continuum *density* in energy, the change of measure must also be included:

$$\rho_E(E) = \sum_{p_i: p_i^2/(2m)=E} \frac{|\tilde{\psi}(p_i)|^2}{|dE/dp|_{p_i}} \quad (1.19)$$

$$= \frac{m}{\sqrt{2mE}} \left[|\tilde{\psi}(\sqrt{2mE})|^2 + |\tilde{\psi}(-\sqrt{2mE})|^2 \right], \quad E > 0. \quad (1.20)$$

The lecture's sum over the two momentum probabilities expresses the essential degeneracy. Equation (1.20) adds the Jacobian needed when probability is quoted per unit energy rather than for an energy bin.

Question from the audience. Should $\hbar$ appear in the energy relation $E = p^2/(2m)$? ^a

Response. No. Momentum and energy retain the familiar nonrelativistic relation. Planck's constant converts those quantities into spatial and temporal phase rates: $k = p/\hbar$ and $\omega = E/\hbar$. It belongs in the wave, not as an extra factor in the dispersion relation.

^aLecture timestamp: 00:47:58–00:48:24.

The physically admissible free states are precisely the suitable superpositions of these modes. For ordinary one-particle wave packets, $\tilde{\psi}$ belongs to $L^2(\mathbb{R})$, so the norm is finite. Solutions that

$$\int_{-\infty}^{\infty} \frac{dp}{\sqrt{2\pi}} \left[e^{i\frac{p}{\hbar}x - i\frac{p^2}{2m\hbar}t} \tilde{\Psi}(p) \right]$$

$$P(q) = \tilde{\Psi}^*(q) \tilde{\Psi}(q)$$

$$E = \frac{p^2}{2m}$$

Figure 1.10: For a given positive energy, the free Hamiltonian admits the two momentum roots $p = \pm\sqrt{2mE}$.

Lecture frame: 00:45:44.

$$\int_{-\infty}^{\infty} \frac{dp}{\sqrt{2\pi}} \left[e^{i\frac{p}{\hbar}x - i\frac{p^2}{2m\hbar}t} \tilde{\Psi}(p) \right]$$

$$P(q) = \tilde{\Psi}^*(q) \tilde{\Psi}(q)$$

$$p_{\pm} = \pm\sqrt{2mE}$$

Figure 1.11: The board collects both momentum amplitudes associated with one free-particle energy.

Lecture frame: 00:48:20.

grow without bound and cannot represent finite probability are not admitted as normalized states, although generalized eigenfunctions remain useful as a basis.

Question from the audience. Does this claim of generality assume that every solution is separable into a function of x times a function of t ? ^a

Response. No. Individual energy–momentum eigenmodes are separable, but a general wave packet is their superposition and is usually not separable. Completeness of the Fourier modes, not separability of the final state, is what makes the integral general.

^aLecture timestamp: 00:49:12–00:50:09.

Question from the audience. May $\tilde{\psi}(p)$ contain an arbitrary momentum-dependent phase? ^a

Response. Yes, and it matters. Its magnitude fixes the momentum probability density, but its phase controls interference among momentum components in position space. Two states with the same $|\tilde{\psi}(p)|^2$ can therefore have very different spatial wave packets.

^aLecture timestamp: 00:50:09–00:50:51.

1.5 Initial Data and Dispersive Evolution

At $t = 0$, the time phase in Equation (1.16) disappears:

$$\psi(x, 0) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp e^{ipx/\hbar} \tilde{\psi}(p). \quad (1.21)$$

Fourier inversion then gives

$$\tilde{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx e^{-ipx/\hbar} \psi(x, 0). \quad (1.22)$$

The factors of $\hbar$ have been kept explicit so that $px/\hbar$ is dimensionless and the two representations have compatible normalization.

The initial-value problem is now constructive:

1. begin with $\psi(x, 0)$;
2. Fourier transform it to obtain $\tilde{\psi}(p, 0)$;
3. multiply each momentum component by $e^{-ip^2 t/(2m\hbar)}$;
4. inverse transform to recover $\psi(x, t)$.

Equivalently,

$$\tilde{\psi}(p, t) = e^{-ip^2 t/(2m\hbar)} \tilde{\psi}(p, 0). \quad (1.23)$$

Question from the audience. Can the time dependence simply be included in a time-dependent momentum wave function? ^a

Response. Yes. Equation (1.23) is exactly that description. Its magnitude remains fixed for a free particle, so momentum probabilities are conserved, while its phase changes at a rate set by $p^2/(2m\hbar)$.

^aLecture timestamp: 00:56:32–00:57:34.

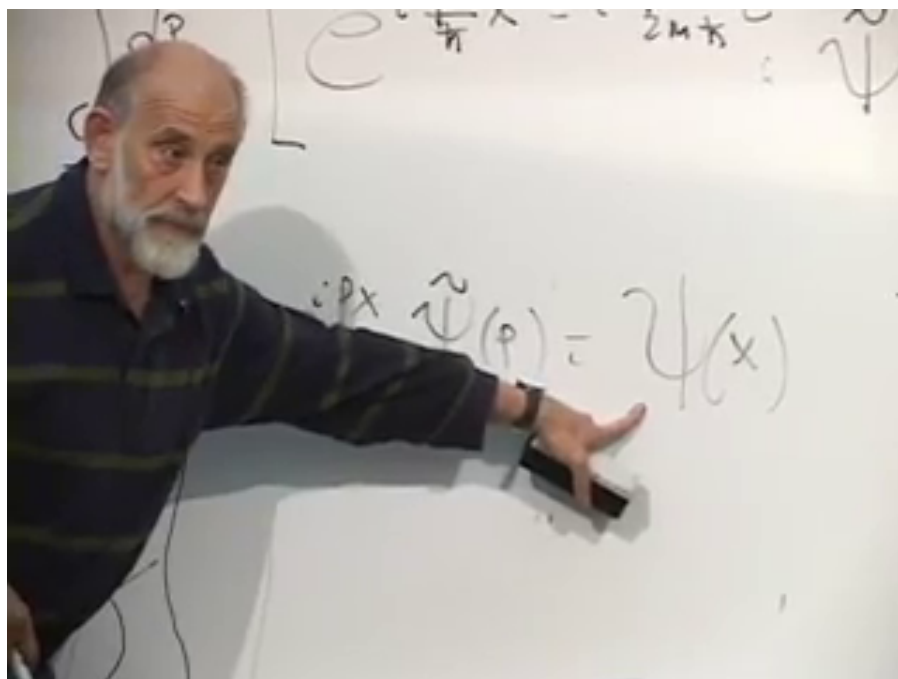


Figure 1.12: Setting $t = 0$ identifies the momentum superposition with the initial position-space wave function.

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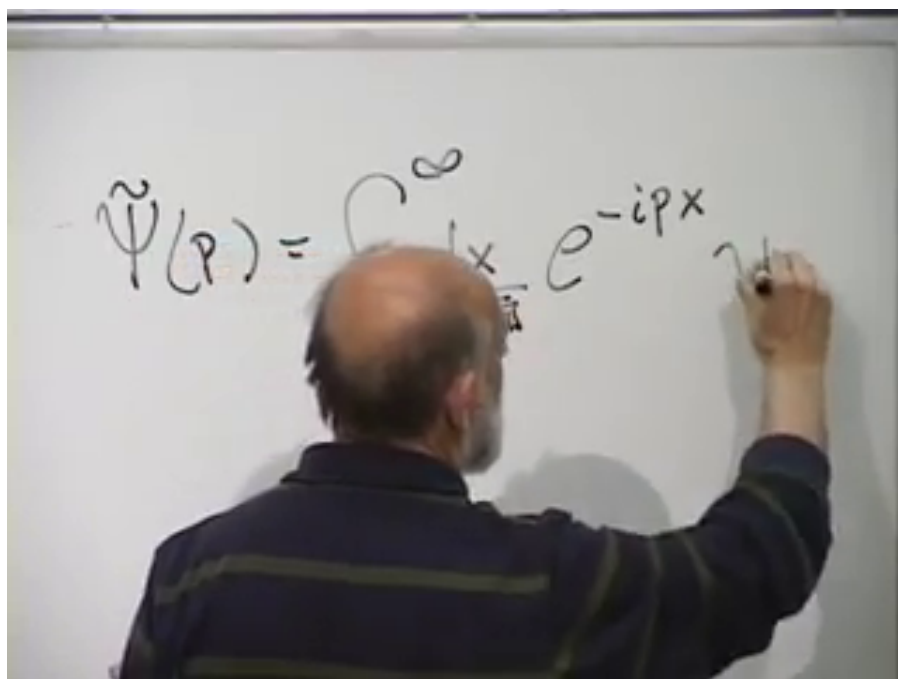


Figure 1.13: The inverse Fourier transform recovers the momentum amplitude from the initial wave function.

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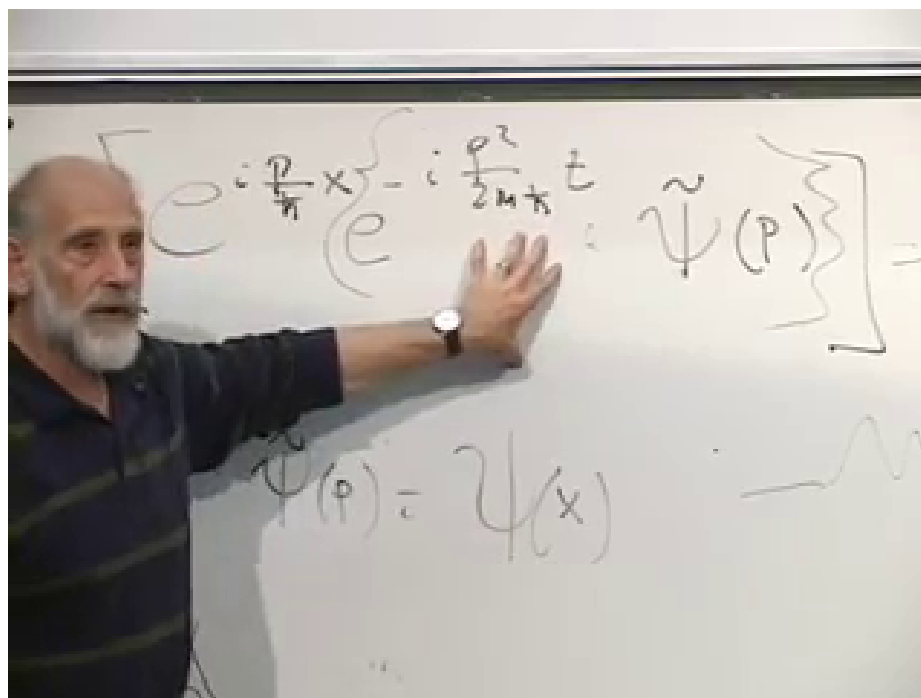


Figure 1.14: Free evolution in momentum space consists of an energy-dependent phase for each Fourier component.

Lecture frame: 00:57:06.

Because the phase is quadratic in p , different components slip relative to one another. A localized packet generally spreads. With $p = \hbar k$ and the conventional positive frequency

$$\omega(k) = \frac{\hbar k^2}{2m}, \quad (1.24)$$

the phase velocity and group velocity are

$$v_{\text{ph}} = \frac{\omega}{k} = \frac{p}{2m}, \quad v_{\text{g}} = \frac{d\omega}{dk} = \frac{p}{m}. \quad (1.25)$$

The group velocity agrees with the classical velocity associated with the central momentum of a narrow packet. The variation of v_{g} over the packet's momentum width is what produces dispersion.

Question from the audience. Is this the distinction between phase velocity and group velocity? ^a

Response. Yes. The crests of an individual Fourier component move with v_{ph} , while the envelope of a narrow packet moves with v_{g} . For the nonrelativistic quadratic dispersion relation they are not equal, and components with different momenta have different group velocities, so a general packet spreads.

^aLecture timestamp: 00:57:35–00:58:17.

$$\begin{aligned}
 &= \langle \psi | \hat{K} | \psi \rangle + \langle \psi | \\
 & i \langle \psi | \hat{H} \hat{K} | \psi \rangle - i \langle \psi | \hat{H} \\
 &= i \langle [\hat{H}, \hat{K}] \rangle
 \end{aligned}$$

Figure 1.15: The two differentiated state vectors combine into the expectation value of a commutator.

Lecture frame: 01:04:50.

Question from the audience. Can the same free-particle equation be used for a photon? ^a

Response. No. Equation (1.5) is the nonrelativistic massive-particle Hamiltonian. A photon obeys $E = c|p|$, not $E = p^2/(2m)$, and has no rest-mass parameter of this kind. Its propagation and polarization require a relativistic field description.

^aLecture timestamp: 01:00:05–01:00:55.

1.6 Commutators and Poisson Brackets

We now step back from the explicit free solution. For an operator $\hat{K}$ that may itself depend on time,

$$\frac{d}{dt} \langle K \rangle = \frac{i}{\hbar} \langle [\hat{H}, \hat{K}] \rangle + \left\langle \frac{\partial \hat{K}}{\partial t} \right\rangle. \quad (1.26)$$

The lecture considers operators with no explicit time dependence, so the last term vanishes. The remaining formula follows by differentiating $\langle \psi | \hat{K} | \psi \rangle$ and applying the Schrödinger equation to both the ket and the bra.

Figure 1.16: Hamilton's equations lead directly to $\dot{K} = \{K, H\}_{\text{PB}}$.

Lecture frame: 01:09:31.

Question from the audience. Does the sign depend on whether the commutator is written as $[\hat{H}, \hat{K}]$ or $[\hat{K}, \hat{H}]$? ^a

Response. Yes, but the two standard forms are identical:

$$\frac{d}{dt}\langle K \rangle = \frac{i}{\hbar} \langle [\hat{H}, \hat{K}] \rangle = \frac{1}{i\hbar} \langle [\hat{K}, \hat{H}] \rangle.$$

Reversing the order changes the sign of the commutator, and $1/i = -i$ changes it back.

^aLecture timestamp: 01:03:11–01:05:17.

The classical law for a phase-space function $K(x, p)$ is

$$\dot{K} = \{K, H\}_{\text{PB}}, \quad (1.27)$$

where

$$\{A, B\}_{\text{PB}} = \frac{\partial A}{\partial x} \frac{\partial B}{\partial p} - \frac{\partial A}{\partial p} \frac{\partial B}{\partial x}. \quad (1.28)$$

The order in Equation (1.27) follows directly from the chain rule and Hamilton's equations:

$$\dot{K} = \frac{\partial K}{\partial x} \dot{x} + \frac{\partial K}{\partial p} \dot{p} \quad (1.29)$$

$$= \frac{\partial K}{\partial x} \frac{\partial H}{\partial p} - \frac{\partial K}{\partial p} \frac{\partial H}{\partial x}. \quad (1.30)$$

The elementary classical identity

$$\{f(x), p\}_{\text{PB}} = f'(x) \quad (1.31)$$

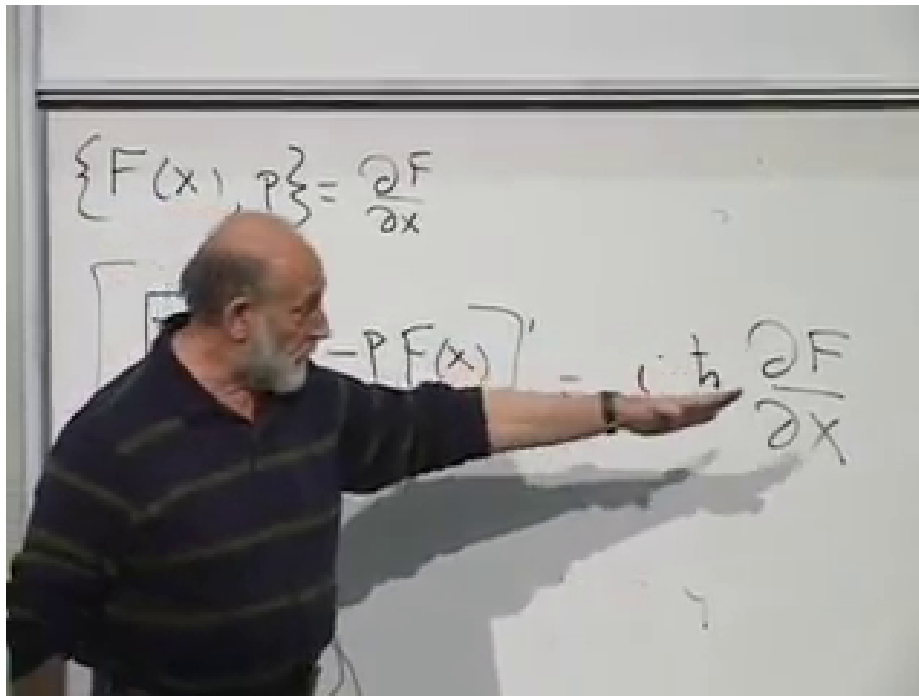


Figure 1.17: Applying the commutator to an arbitrary wave function exposes the derivative $f'(\hat{x})$.

Lecture frame: 01:19:12.

has an exact quantum counterpart. Act with the commutator on an arbitrary test wave function:

$$[f(\hat{x}), \hat{p}]\psi = f(x) \left(-i\hbar \frac{d\psi}{dx} \right) + i\hbar \frac{d}{dx} (f(x)\psi(x)) \quad (1.32)$$

$$= i\hbar f'(x)\psi(x). \quad (1.33)$$

Because this holds for every wave function in the common operator domain,

$$\boxed{[f(\hat{x}), \hat{p}] = i\hbar f'(\hat{x})}. \quad (1.34)$$

In the momentum representation the reciprocal identity is

$$[\hat{x}, g(\hat{p})] = i\hbar g'(\hat{p}), \quad (1.35)$$

and the special case $f(x) = x$ yields

$$[\hat{x}, \hat{p}] = i\hbar. \quad (1.36)$$

These examples motivate the correspondence

$$[\hat{A}, \hat{B}] \longleftrightarrow i\hbar \widehat{\{A, B\}_{\text{PB}}}. \quad (1.37)$$

Both brackets are antisymmetric,

$$[A, B] = -[B, A], \quad \{A, B\}_{\text{PB}} = -\{B, A\}_{\text{PB}}, \quad (1.38)$$

and both act as derivations on products:

$$[AB, C] = A[B, C] + [A, C]B, \quad (1.39)$$

$$\{AB, C\}_{\text{PB}} = A\{B, C\}_{\text{PB}} + \{A, C\}_{\text{PB}}B. \quad (1.40)$$

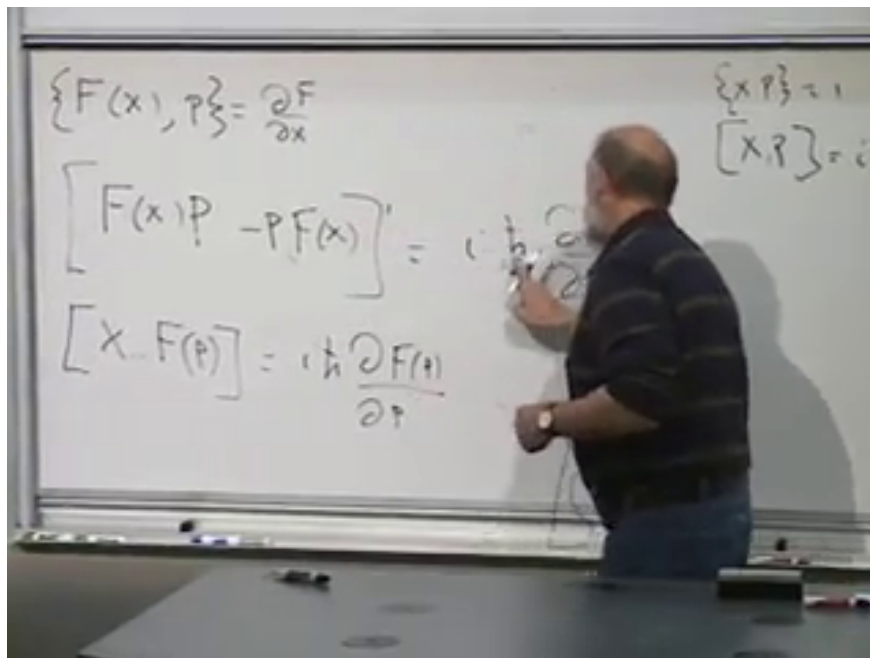


Figure 1.18: The canonical commutator and Poisson bracket display the basic quantum–classical correspondence.

Lecture frame: 01:21:52.

For classical functions the order of the ordinary factors is immaterial. For operators it is not, which is why the placement of B in Equation (1.39) must be preserved.

Question from the audience. In the commutator product rule, should the second term be $B[A, C]$ or $[A, C]B$? ^a

Response. It is $[A, C]B$. Expanding $ABC - CAB$ and adding and subtracting ACB gives

$$ABC - ACB + ACB - CAB = A[B, C] + [A, C]B.$$

The classroom correction matters because the factors need not commute.

^aLecture timestamp: 01:29:31–01:30:55.

The correspondence in Equation (1.37) is exact for the canonical elementary identities just derived, but it is not a universal replacement rule for arbitrary functions. A classical monomial such as x^3p has several inequivalent operator orderings. Symmetrization often provides a natural choice, yet higher-order $\hbar$ corrections remain in general. The classical bracket is the leading correspondence limit, not a license to reverse or rearrange arbitrary operator strings.

1.7 A Potential and Ehrenfest's Theorem

Add a position-dependent potential:

$$\hat{H} = \frac{\hat{p}^2}{2m} + U(\hat{x}). \quad (1.41)$$

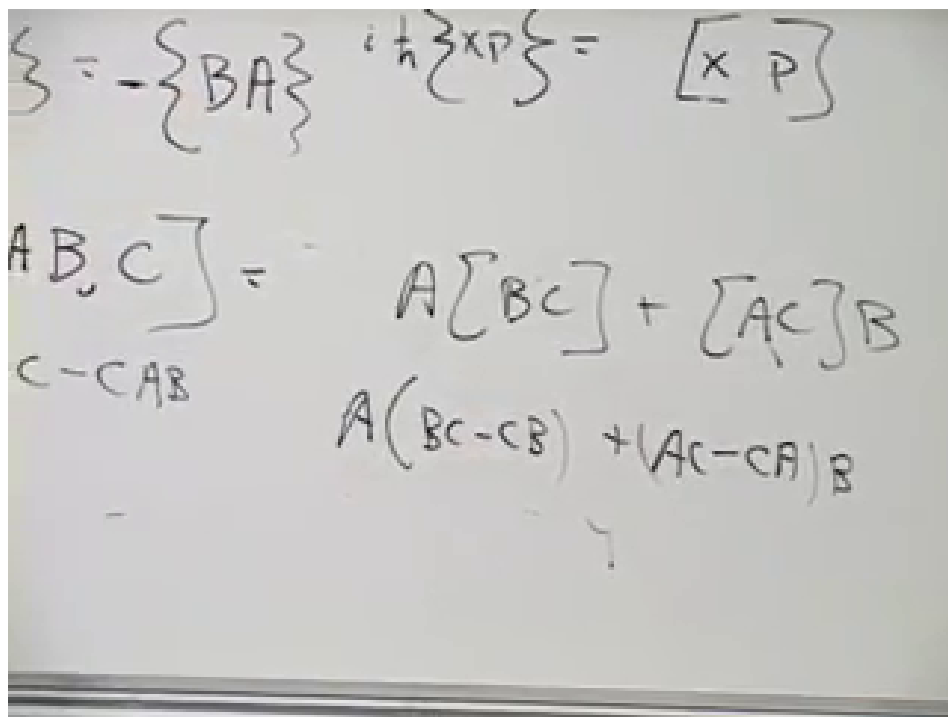


Figure 1.19: The product rules for Poisson brackets and commutators, with operator order retained in the quantum identity.

Lecture frame: 01:29:55.

In position space, $U(\hat{x})$ acts by multiplication, and the Schrödinger equation becomes

$$i\hbar \frac{\partial \psi(x, t)}{\partial t} = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + U(x) \right] \psi(x, t). \quad (1.42)$$

The associated force is

$$F(x) = -U'(x). \quad (1.43)$$

Question from the audience. Does the momentum operator itself now depend on time? ^a

Response. In the Schrödinger picture, $\hat{p} = -i\hbar \partial_x$ has no explicit time dependence. The state changes, and therefore its momentum distribution and expectation value can change. One may instead transfer time dependence to operators in the Heisenberg picture, but that is a different representation of the same physics.

^aLecture timestamp: 01:35:01–01:35:40.

We now ask how the center of a wave packet moves. First set $\hat{K} = \hat{x}$ in Equation (1.26). Since $[U(\hat{x}), \hat{x}] = 0$,

$$\frac{d}{dt} \langle x \rangle = \frac{i}{\hbar} \left\langle \left[\frac{\hat{p}^2}{2m}, \hat{x} \right] \right\rangle. \quad (1.44)$$

The product rule and $[\hat{p}, \hat{x}] = -i\hbar$ give

$$[\hat{p}^2, \hat{x}] = \hat{p}[\hat{p}, \hat{x}] + [\hat{p}, \hat{x}]\hat{p} \quad (1.45)$$

$$= -2i\hbar \hat{p}. \quad (1.46)$$

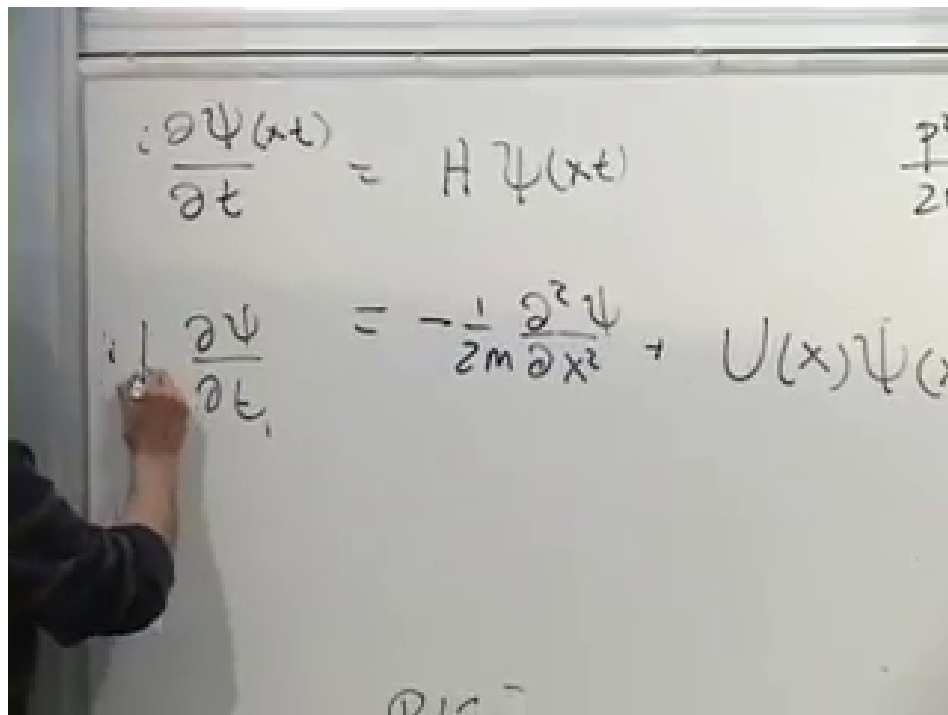


Figure 1.20: The Schrödinger equation for kinetic energy plus a position-dependent potential.

Lecture frame: 01:34:48.

Consequently,

$$\boxed{\frac{d}{dt}\langle x \rangle = \frac{\langle p \rangle}{m}}. \quad (1.47)$$

Next choose $\hat{K} = \hat{p}$. The kinetic term commutes with momentum, while Equation (1.34) gives $[U(\hat{x}), \hat{p}] = i\hbar U'(\hat{x})$. Therefore

$$\frac{d}{dt}\langle p \rangle = \frac{i}{\hbar} \langle [U(\hat{x}), \hat{p}] \rangle \quad (1.48)$$

$$= -\langle U'(\hat{x}) \rangle = \langle F(\hat{x}) \rangle. \quad (1.49)$$

Combining the two exact identities,

$$m \frac{d^2}{dt^2} \langle x \rangle = -\langle U'(\hat{x}) \rangle. \quad (1.50)$$

This resembles Newton's equation, but an important distinction remains:

$$\langle U'(\hat{x}) \rangle \neq U'(\langle x \rangle) \quad \text{in general.} \quad (1.51)$$

If the packet is narrow compared with the length scale over which the force changes, then

$$\langle U'(\hat{x}) \rangle \approx U'(\langle x \rangle), \quad (1.52)$$

and

$$m \frac{d^2}{dt^2} \langle x \rangle \approx F(\langle x \rangle). \quad (1.53)$$

$$\frac{d}{dt} \langle X \rangle = \frac{i}{\hbar} \langle \left[\frac{p^2}{2m}, X \right] \rangle$$

$$\frac{d}{dt} \langle X \rangle = \frac{1}{m} \langle p \rangle$$

Figure 1.21: The first Ehrenfest equation: average velocity equals average momentum divided by mass.

Lecture frame: 01:42:18.

$$\frac{d}{dt} \langle P \rangle = \frac{i}{\hbar} \langle \left[\frac{p^2}{2m} + V(x), P \right] \rangle$$

$$\frac{d}{dt} \langle P \rangle = - \left\langle \frac{\partial V}{\partial x} \right\rangle = F$$

Figure 1.22: The second Ehrenfest equation: the mean momentum changes by the expectation value of the force.

Lecture frame: 01:45:39.

For potentials of at most quadratic degree, $U'(x)$ is at most linear, so the replacement in Equation (1.52) is exact for the first moments even when the packet is broad. For a general nonlinear potential, packet width and higher moments matter.

Question from the audience. Are all the apparent Poisson-bracket and commutator analogies exact, and is $\langle F \rangle$ the quantum version of the resultant force? ^a

Response. The elementary canonical relations and the two Ehrenfest equations above are exact. A general replacement of every Poisson bracket by a commutator is only a correspondence rule because operator ordering can matter. In one dimension, $\langle F(\hat{x}) \rangle$ is the probability weighted average force acting across the packet. It equals the force at the average position only under the additional conditions just stated.

^aLecture timestamp: 01:46:36–01:47:35.

This is the promised order of explanation. Quantum mechanics supplies the fundamental evolution. Classical motion emerges when the observables of interest vary little across the state, so that averages close approximately on themselves. We do not first assume a classical particle and then decorate it with a wave; we derive the motion of a classical-looking center from the quantum wave packet.

1.8 What Has Been Established

The historical crisis taught us not to confuse a successful approximation with a fundamental mechanism. With that lesson in place, the free Hamiltonian $\hat{p}^2/(2m)$ led to the Schrödinger equation, plane waves, and the dispersion relation $E = p^2/(2m)$. Their Fourier superpositions gave every normalizable free state and a direct procedure for evolving arbitrary initial data.

The second half of the lecture exposed the algebra beneath the quantum–classical relation. Commutators govern expectation values in the same structural position that Poisson brackets govern classical observables. For a particle in a potential, that parallel produces the exact Ehrenfest equations

$$\frac{d}{dt}\langle x \rangle = \frac{\langle p \rangle}{m}, \quad \frac{d}{dt}\langle p \rangle = -\langle U'(\hat{x}) \rangle. \quad (1.54)$$

Only after these quantum statements are established do we take the narrow-packet, smooth-force limit and recover Newton’s equation for the packet’s center.